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Experiment M5 Report
Normal and Oblique Wave Incidence

ECE 331 – Electromagnetic Waves (Microwaves Laboratory)

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G2

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Objective

The objective of this experiment is to study the propagation of plane waves when they encounter a boundary between two different media. This report analyzes two specific cases:

1. **Normal Incidence:** Verifying the reflection and transmission coefficients, Standing Wave Ratio (SWR), and power conservation for both lossless and lossy media.
2. **Oblique Incidence:** Analyzing wave behavior at general angles, including parallel and perpendicular polarization, and verifying phenomena such as the Brewster angle and Critical angle.

Theoretical Background

When a plane wave impinges on a boundary between two media, it is divided into a transmitted wave and a reflected wave.

1. Normal Incidence

For a wave striking the boundary normally (z=0), the reflection coefficient and transmission coefficient are determined by the intrinsic impedances of the media:

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \quad \text{and} \quad T = \frac{2\eta_2}{\eta_2 + \eta_1}$$

The Standing Wave Ratio (SWR) in Medium 1 is defined as:

$$SWR = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

Power conservation dictates that the incident power equals the sum of reflected and transmitted power.

2. Oblique Incidence

For oblique incidence, the reflection and transmission depend on the wave polarization (Perpendicular or Parallel) and the angle of incidence.

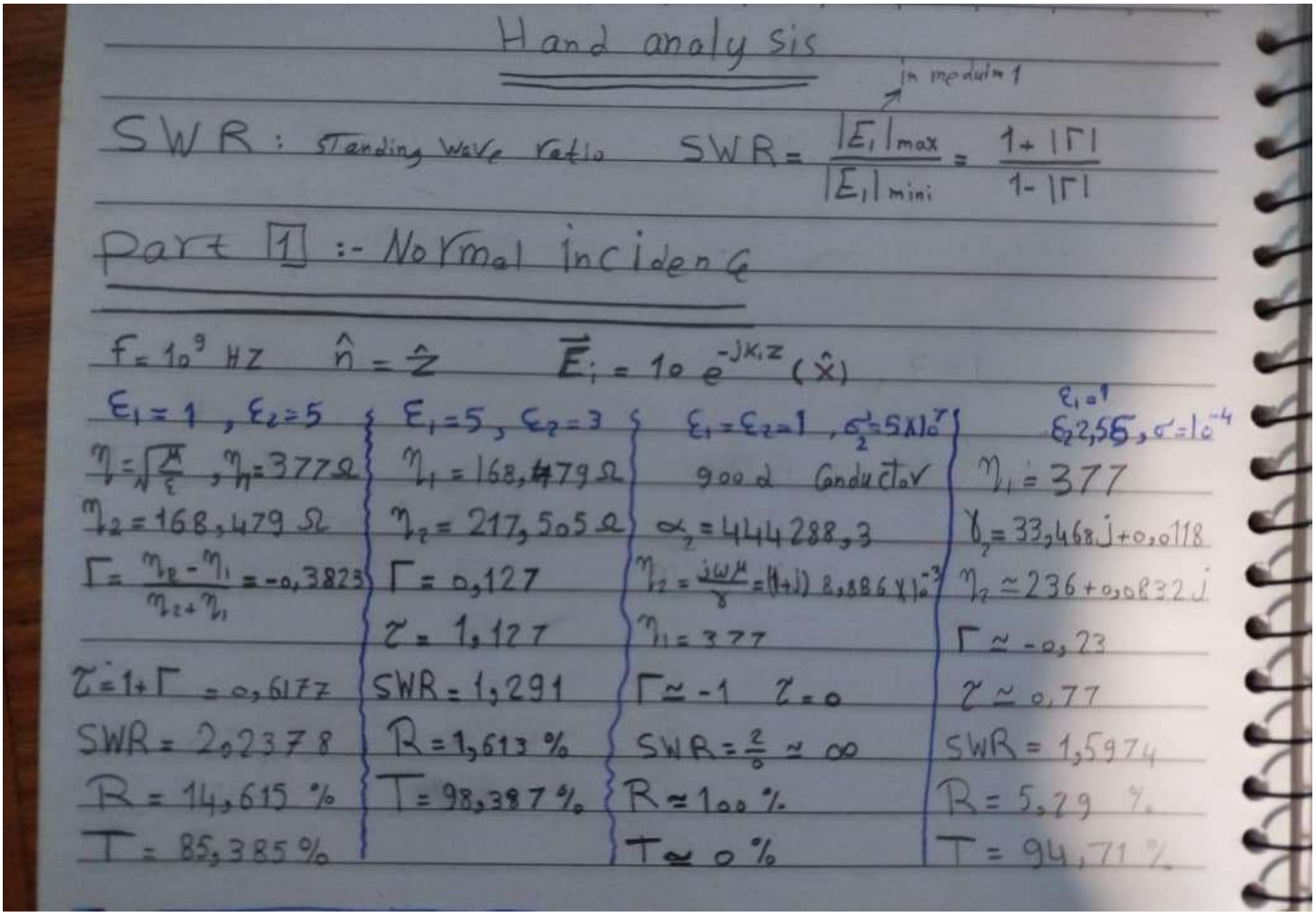
Snell's Laws govern the relationship between the angles of reflection and transmission.

Brewster Angle: The angle where total transmission occurs (no reflection). This exists only for parallel polarization in non-magnetic media.

Critical Angle: The angle of incidence above which total internal reflection occurs (only when (epsilon_1 > epsilon_2).

Part 1: Normal Incidence

1 Hand Analysis Results: Below are the calculated values for the 1 GHz plane wave incident on the boundary with an electric field amplitude of 10 V/m

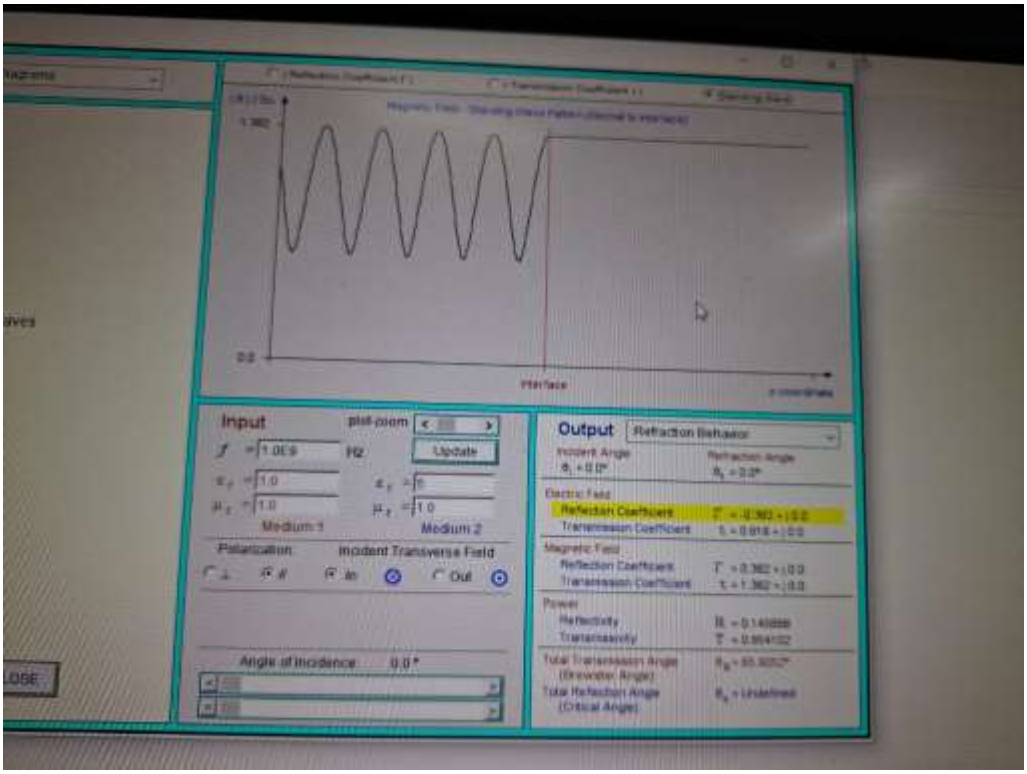


2 Simulation Results The following figures show the standing wave patterns generated

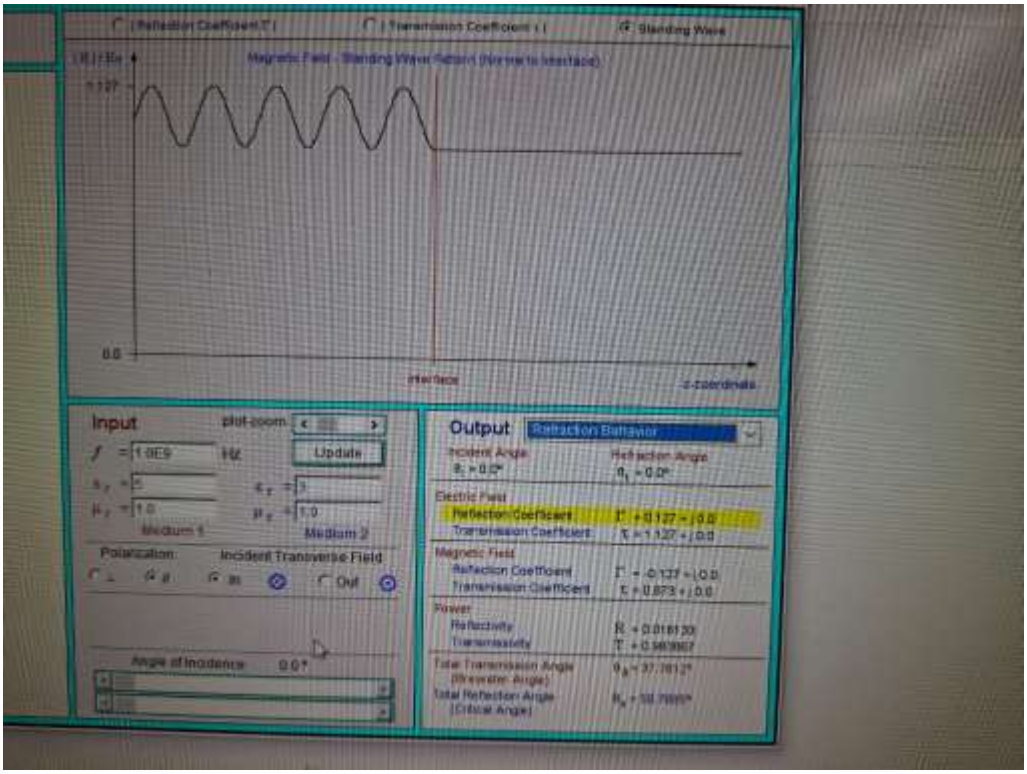
Input Settings: Polarization = //, Angle = 0

Part 1: Lossless medium

Case (i): ε1 = 1, ε2 = 5

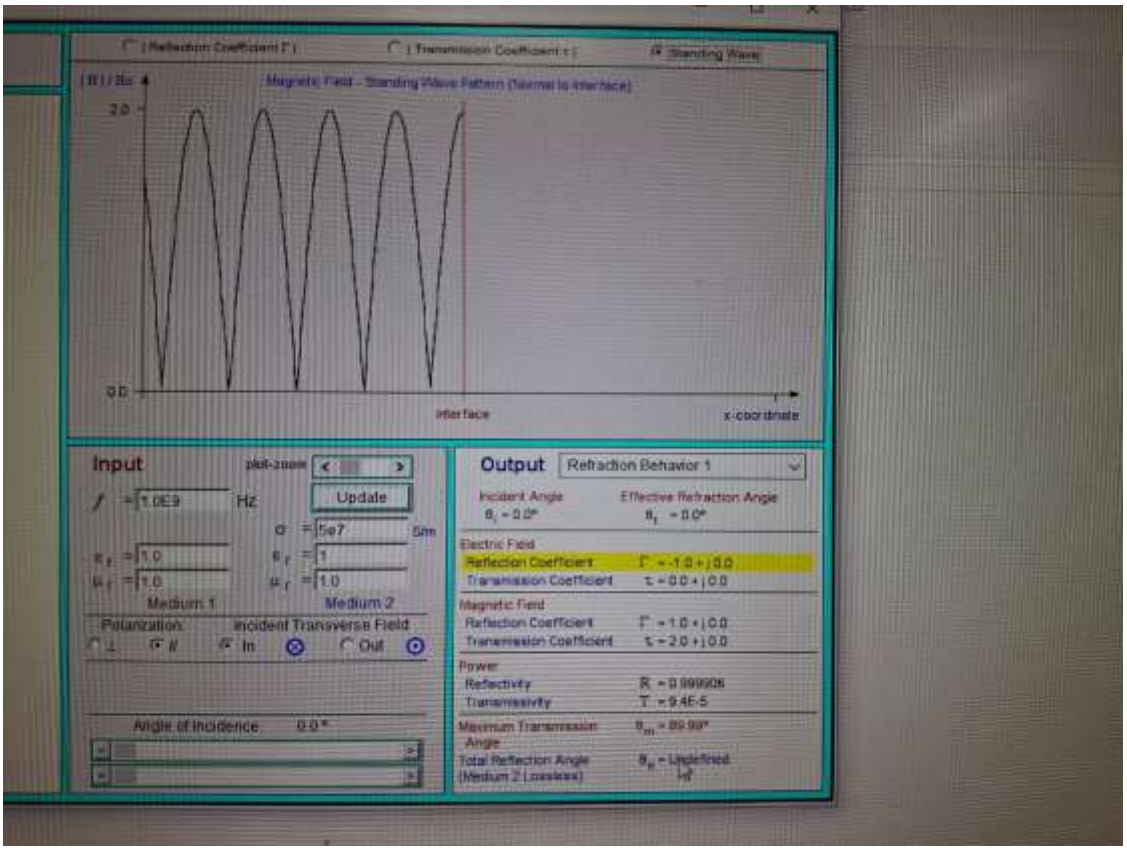


Case (ii): ε1 = 5, ε2 = 3

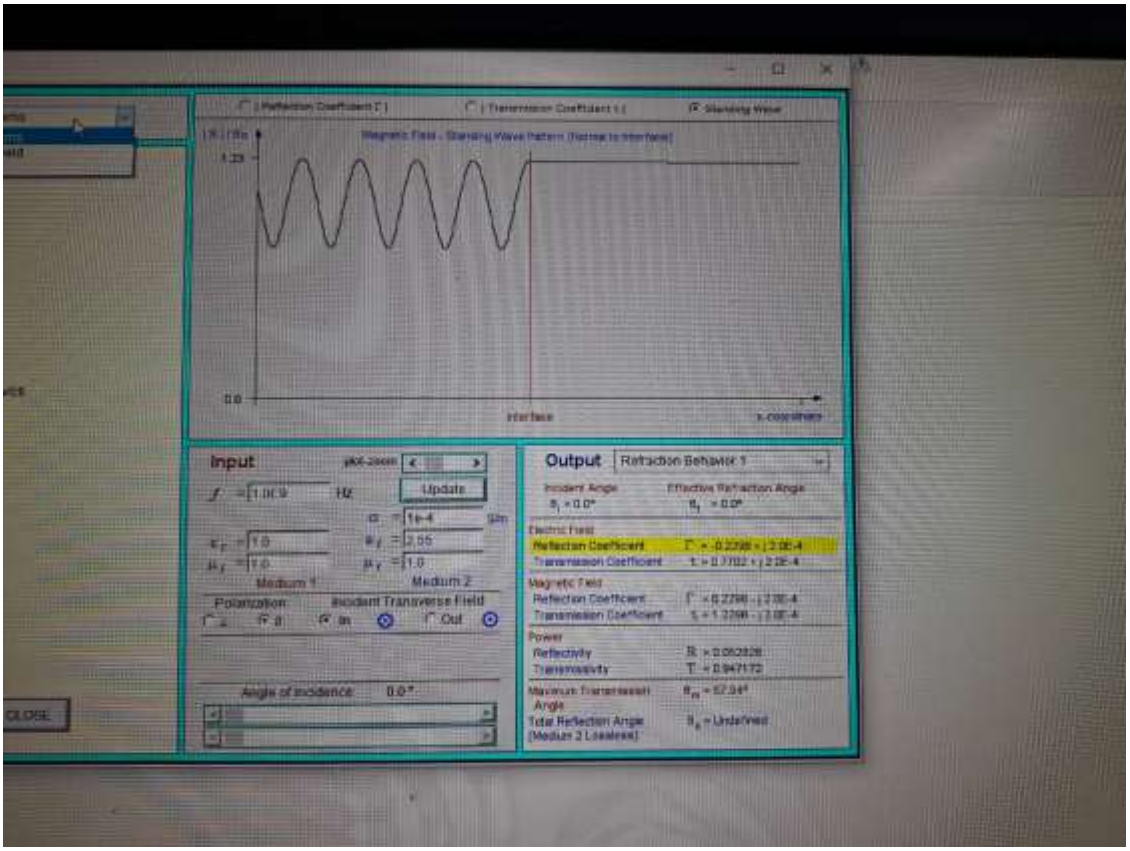


Part 2: Lossy medium

Case (iii): ε1 = 1, ε2 = 1, σ = 5x107 S/m



Case (iv): ε1 = 1, ε2 = 2.55, σ = 10-4 S/m

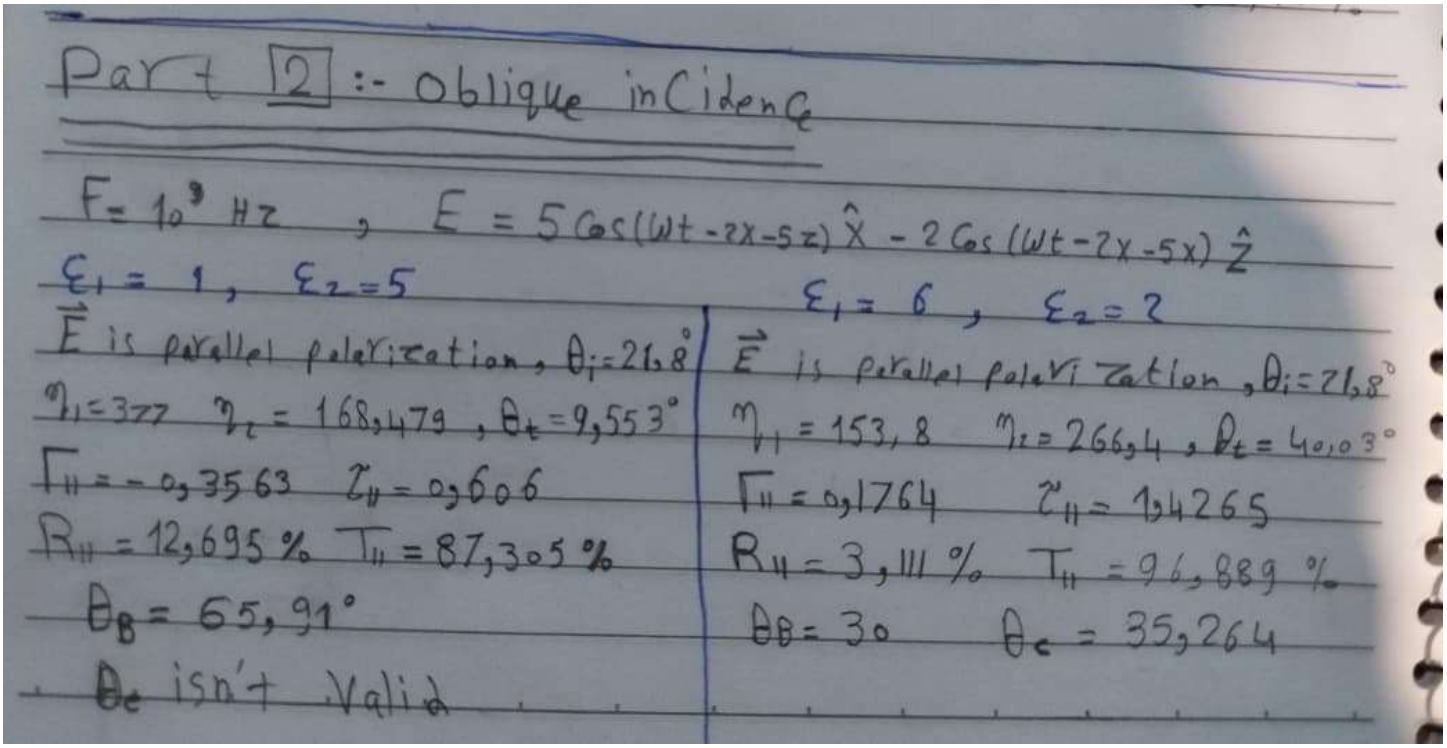


Part 2: Oblique Incidence

The expression of the electric field is

$E = \hat{x}\,5\cos(\omega t - 2x - 5z) - \hat{z}\,2\cos(\omega t - 2x - 5z)$

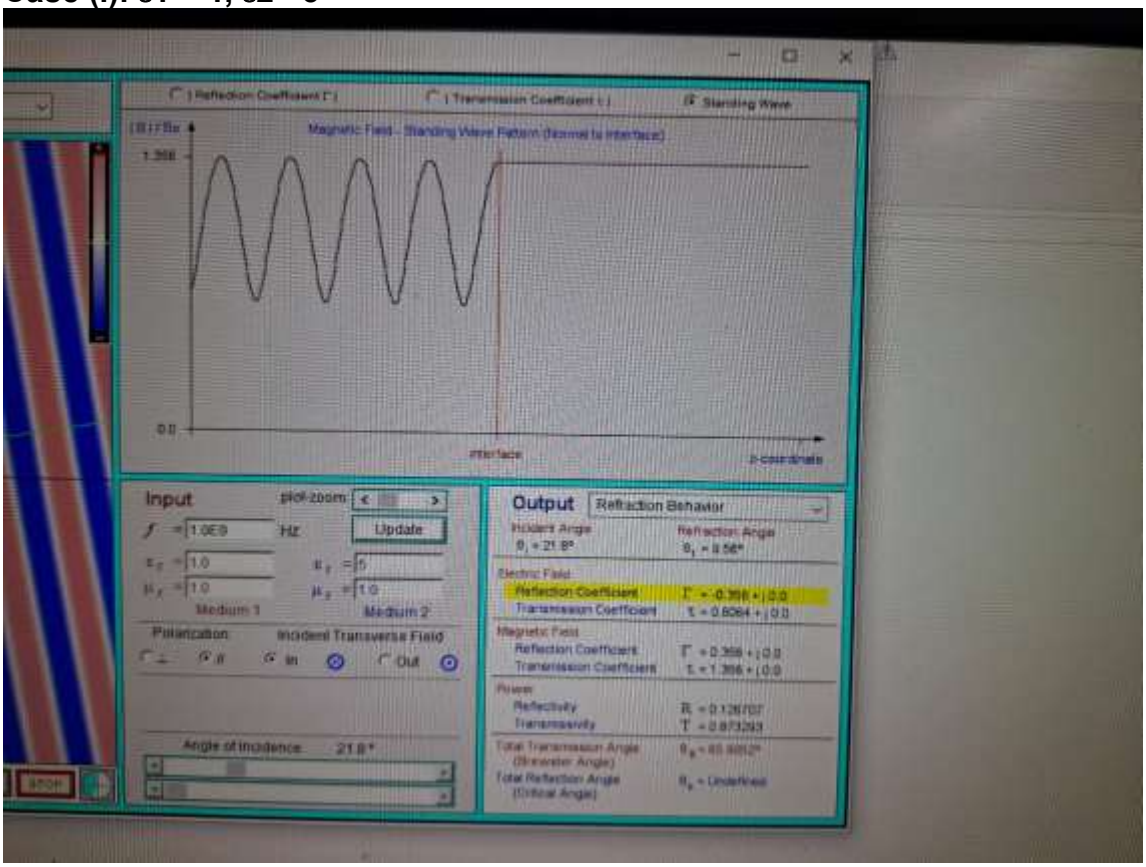
1 Hand Analysis Results: Below are the calculated values for the 1 GHz plane wave propagating in medium 1.

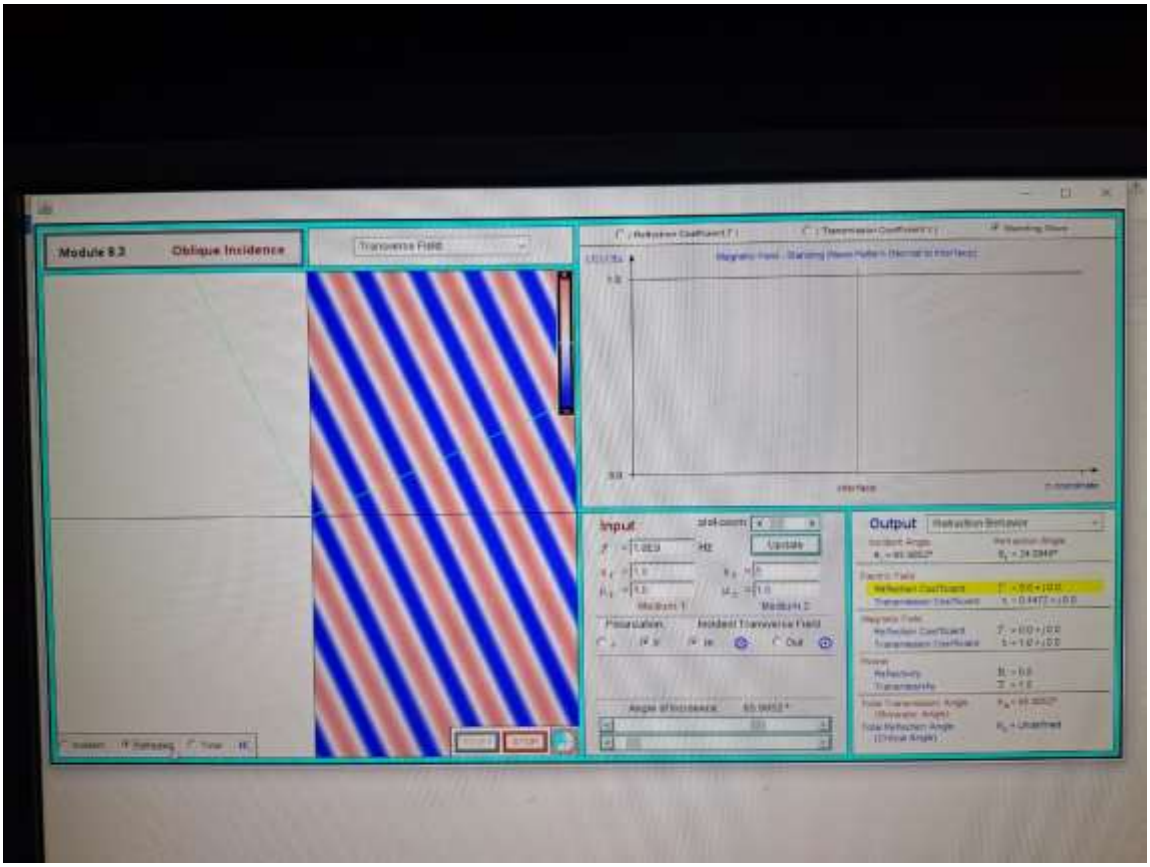


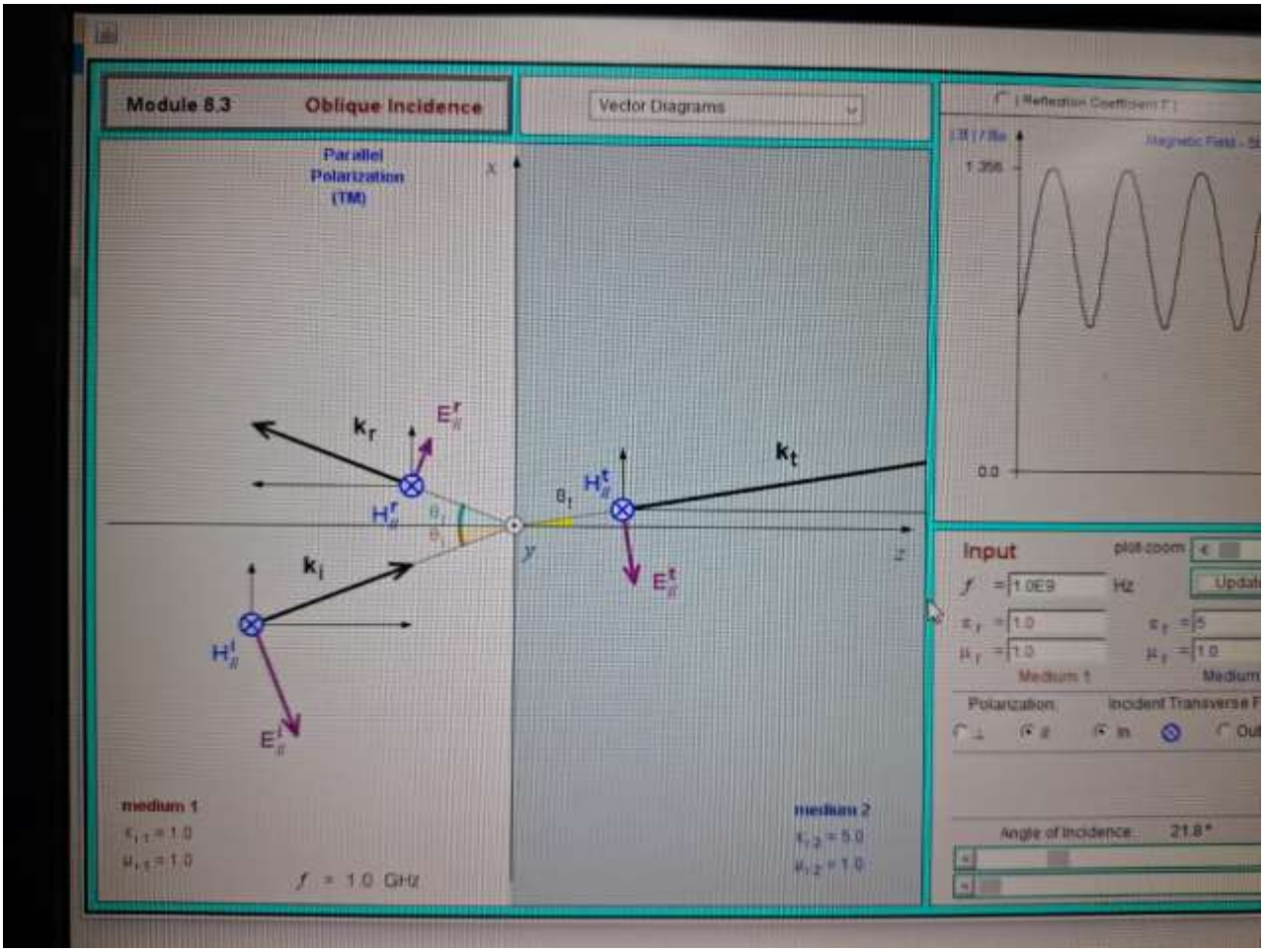
2 Simulation Results The following figures show the standing wave patterns generated using

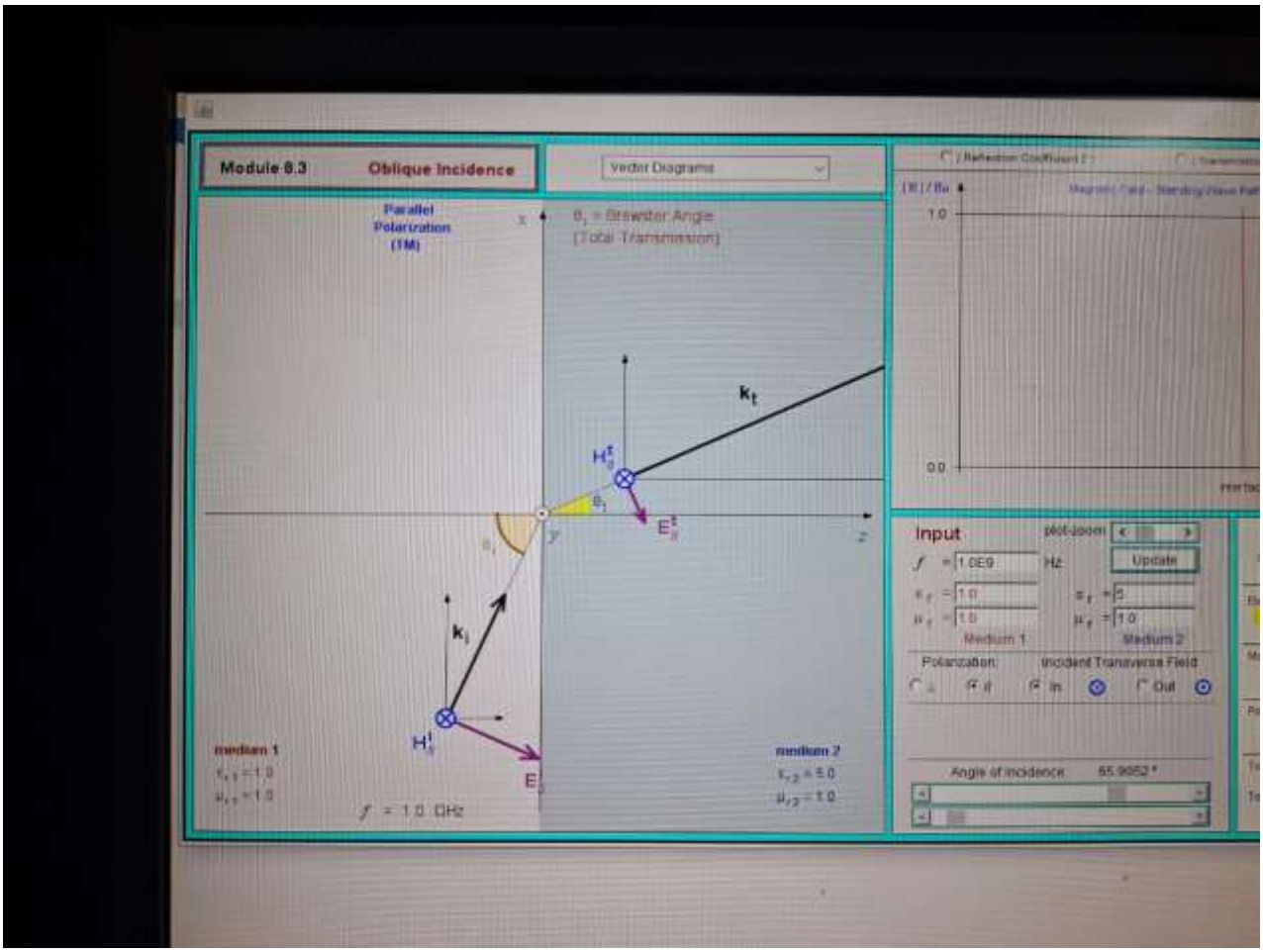
Input Settings: Polarization = //, Angle = 0

Case (i): ε1 = 1, ε2= 5

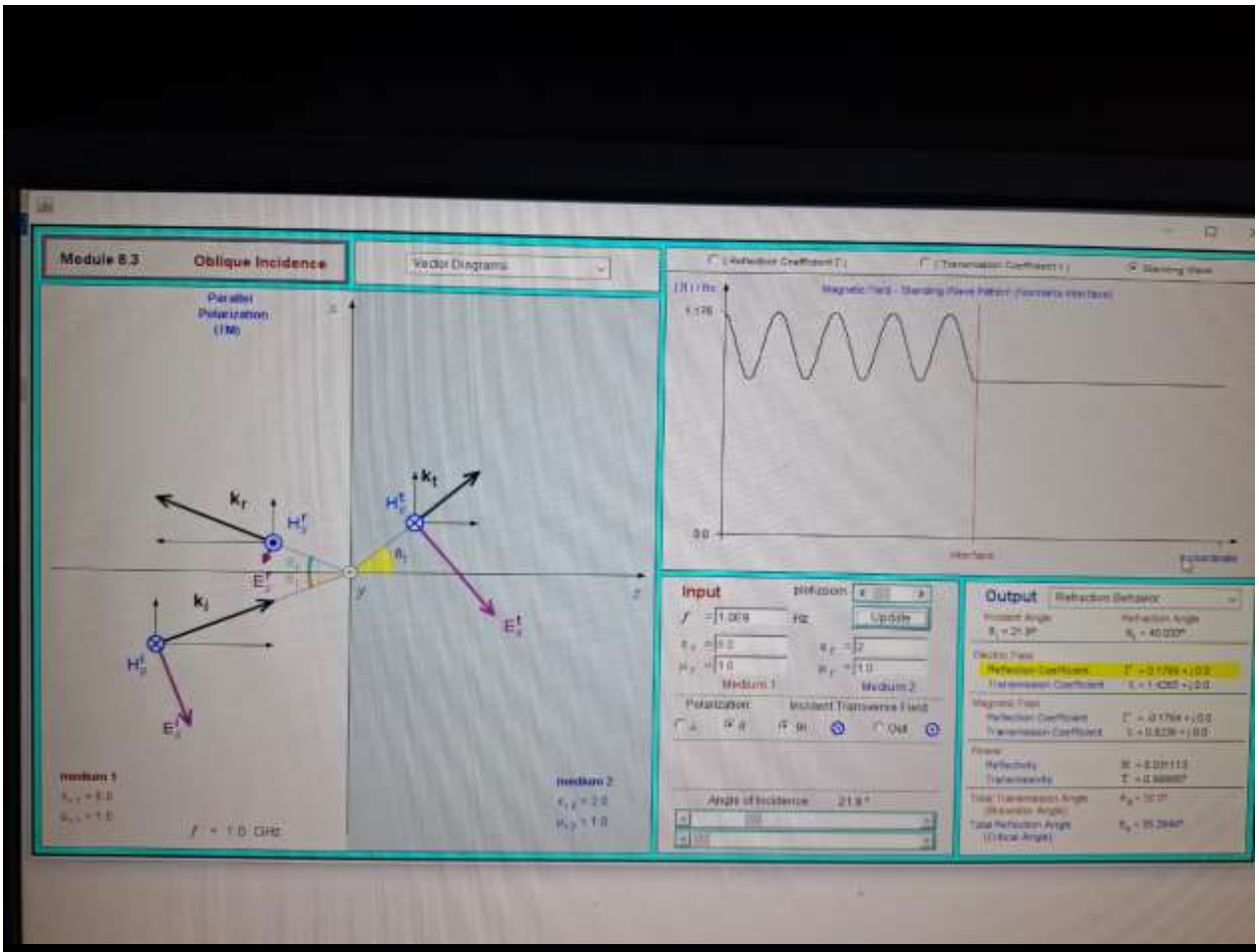


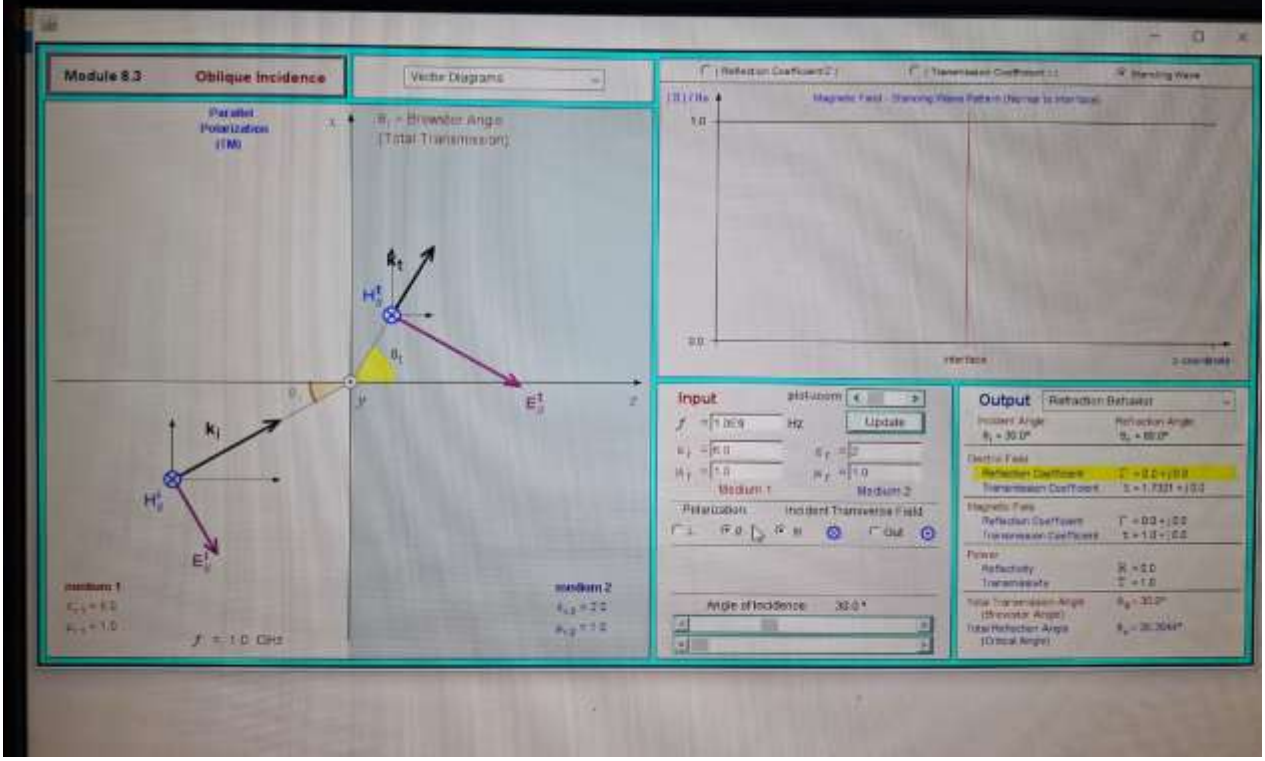


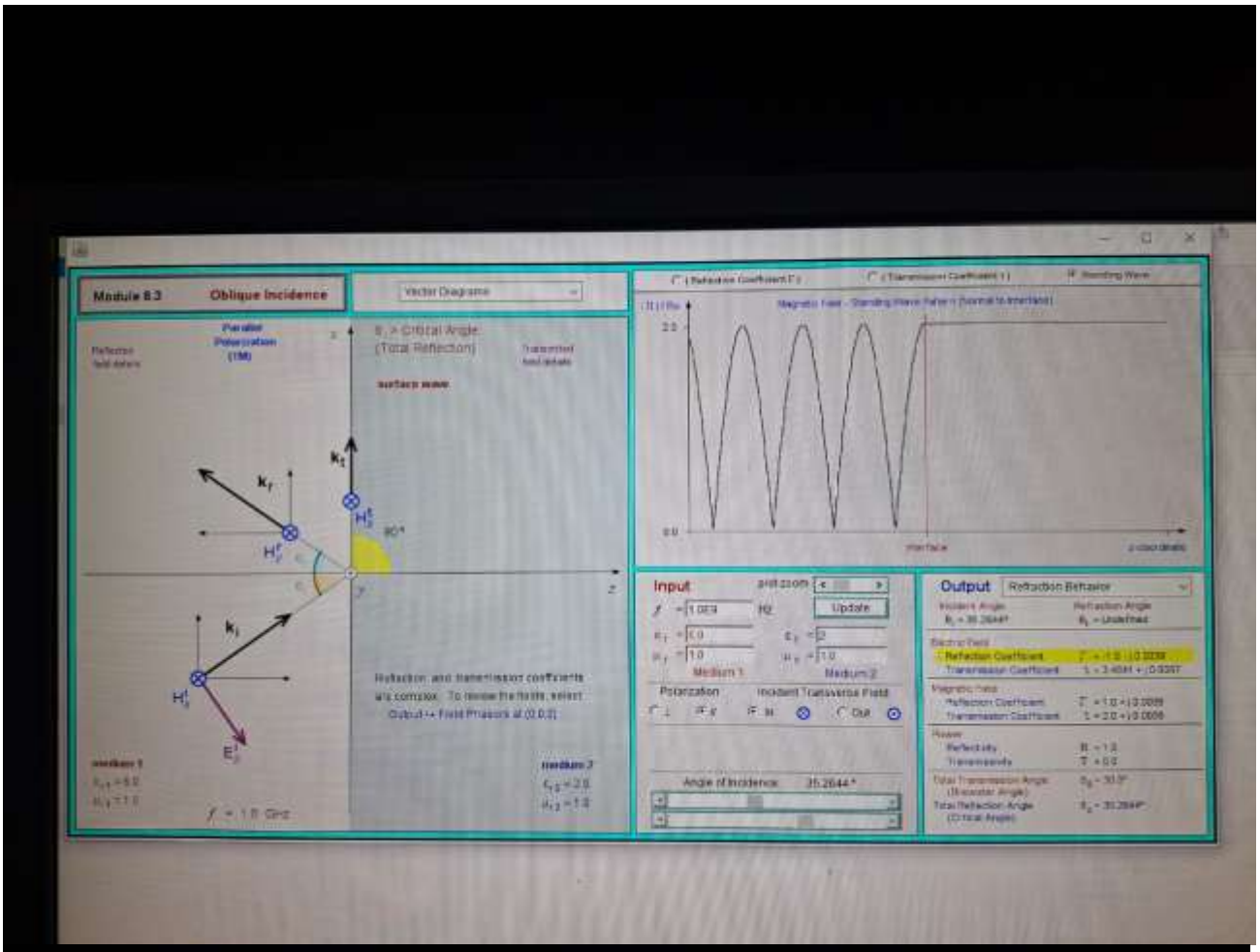

Module 8.3
Oblique Incidence
Vector Diagrams
Parallel
Polarization
(TM)
Input
plot-zoom
Update
Hz
Medium 1
Medium 2
Polarization
Incident Transverse F
Angle of Incidence
21.8°
medium 1
medium 2
f = 1.0 GHz

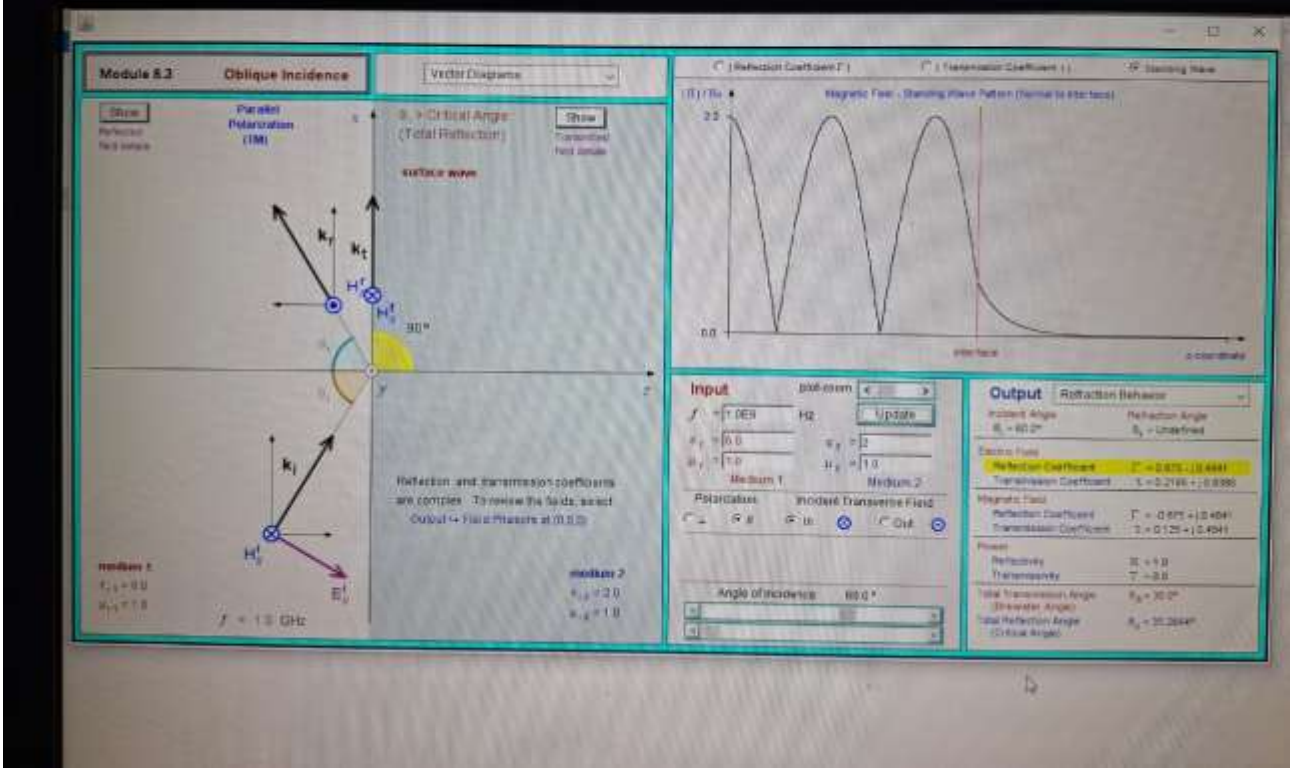

Module 8.3
Oblique Incidence
Vector Diagrams
Parallel
Polarization
(TM)
θi = Brewster Angle
[Total Transmission]
Input
plot-zoom
Update
Hz
Medium 1
Medium 2
Polarization
Incident Transverse Field
Angle of Incidence
65.9952°
medium 1
medium 2
f = 1.0 GHz

Case (ii): ε1 = 6, ε2 = 2






Module 8.3
Oblique Incidence
Vector Diagrams
Parallel Polarization (TM)
θi > Critical Angle (Total Reflection)
surface wave
Reflection and transmission coefficients are complex. To review the fields, select Output ↦ Field Phasors at (0,0,0)
medium 1
medium 2
f = 1.0 GHz
Input
pan zoom
Update
Medium 1
Medium 2
Polarization
Incident Transverse Field
Angle of Incidence
35.2644°
Output
Refraction Behavior
Incident Angle
Refraction Angle
Reflection Coefficient
Transmission Coefficient
Power
Reflectivity
Transmissivity
Total Transmission Angle (Brewster Angle)
Total Reflection Angle (Critical Angle)


Module 8.3
Oblique Incidence
Vector Diagrams
Show
Parallel Polarization (TM)
θi > Critical Angle (Total Reflection)
surface wave
Reflection and transmission coefficients are complex. To review the fields, select Output ↦ Field Phasors at (0,0,0)
medium 1
medium 2
f = 1.0 GHz
Input
pan zoom
Update
Medium 1
Medium 2
Polarization
Incident Transverse Field
Angle of Incidence
60.0°
Output
Refraction Behavior
Incident Angle
Refraction Angle
Reflection Coefficient
Transmission Coefficient
Power
Reflectivity
Transmissivity
Total Transmission Angle (Brewster Angle)
Total Reflection Angle (Critical Angle)